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TOPIC:

**Assessing the Effectiveness of Birth Plans in Improving Maternal
Satisfaction among Women at Sunyani Municipal Hospital**

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Table Of Contents

CHAPTER ONE	4
INTRODUCTION	4
1.1 Background to the Study	4
1.2 Problem Statement.....	7
1.3 Objectives of the Study.....	9
1.3.1 General Objective	9
1.3.2 Specific Objectives	9
1.4 Research Questions.....	9
1.5 Significance of the Study.....	10
1.6 Organization of Chapters	10
CHAPTER TWO.....	11
LITERATURE REVIEW	11
2.1 Introduction	11
2.2 Key Topics for First Specific Objective: Awareness and Utilization of Birth Plans ..	13
2.2.1 Concept and Evolution of Birth Plans	13
2.2.2 Awareness of Birth Plans among Pregnant Women.....	14
2.2.3 Utilization of Birth Plans in Clinical Practice	15
2.2.4 Barriers to Awareness and Utilization	16
2.3 Key Topics for Second Specific Objective: Relationship Between Birth Plans and Maternal Satisfaction.....	17
2.3.1 Concept of Maternal Satisfaction	17
2.3.2 Theoretical Link Between Birth Plans and Satisfaction	18
2.3.3 Empirical Evidence Supporting Positive Outcomes.....	18
2.3.4 Contradictory Evidence	19
2.3.5 Importance of Expectation Management.....	19
2.4 Factors Influencing the Effectiveness of Birth Plans	19

2.4.1 Healthcare Provider Attitudes and Practices	19
2.4.2 Communication and Shared Decision-Making.....	20
2.4.3 Health System Factors	20
2.4.4 Sociocultural Factors	20
2.4.5 Educational Level and Awareness	21
2.4.6 Flexibility of Birth Plans	21
2.5 Conclusion	21
CHAPTER THREE.....	22
METHODOLOGY	22
3.1 Background of the Study Area.....	22
3.2 Study Design and Type	23
3.3 Study Population.....	24
3.4 Sample Size and Sampling Technique.....	26
3.4.1 Sample Size Determination	26
3.4.2 Sampling Technique	26
3.5 Study Variables	27
3.6 Data Collection Tool and Technique	28
3.7 Data Processing and Analysis.....	29
3.8 Limitations of the Study	30
3.9 Ethical Considerations	30
References	31

CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

Maternal health continues to be a major global public health concern, particularly in low- and middle-income countries where disparities in access to quality healthcare services remain pronounced. Over the past few decades, international health initiatives have focused heavily on reducing maternal mortality rates, leading to significant improvements worldwide. According to the World Health Organization (WHO), global maternal mortality declined by approximately 38% between 2000 and 2017; however, sub-Saharan Africa still accounts for nearly two-thirds of maternal deaths globally (World Health Organization [WHO], 2019). While these efforts have been instrumental in improving survival outcomes, there is increasing recognition that maternal health extends beyond survival to include the quality of care and women's experiences during pregnancy, labour, and childbirth.

In response to this shift, global health discourse has increasingly emphasized respectful maternity care and positive childbirth experiences as essential components of quality maternal healthcare. The WHO highlights that a positive childbirth experience is one that fulfills or exceeds a woman's prior personal and sociocultural beliefs and expectations, including giving birth to a healthy baby in a clinically safe environment with continuous emotional support, effective communication, and respectful care (Oladapo et al., 2018). This paradigm shift reflects a broader movement toward patient-centered care, where women are not merely recipients of care but active participants in decision-making processes affecting their health and well-being.

One of the key indicators used to assess the quality of maternal healthcare services is maternal satisfaction. Maternal satisfaction refers to women's evaluation of the care they receive during pregnancy, labour, delivery, and the postpartum period. It is a multidimensional concept influenced by several factors, including interpersonal relationships with healthcare providers, the physical environment of healthcare facilities, accessibility of services, technical competence of providers, and the degree of involvement in decision-making (Srivastava et al., 2015). Research indicates that women who perceive their care as respectful, supportive, and inclusive are more likely to report higher levels of satisfaction (Chabbert et al., 2021).

Maternal satisfaction is not only important as an outcome in itself but also has significant implications for healthcare utilization and maternal and neonatal outcomes. Women who have positive childbirth experiences are more likely to utilize skilled birth services in subsequent pregnancies, adhere to postnatal care recommendations, and recommend facility-based delivery to others (Bohren et al., 2014). Conversely, negative experiences such as disrespect, neglect, or lack of communication can discourage women from seeking facility-based care, thereby increasing the risk of adverse maternal and neonatal outcomes.

In recent years, birth plans have emerged as an important strategy for promoting maternal satisfaction and enhancing the quality of maternity care. A birth plan is a written document prepared by a pregnant woman, often with the guidance of a midwife or healthcare provider, outlining her preferences for labour and delivery. These preferences may include choices related to pain management, labour positioning, presence of a birth companion, mode of delivery, interventions such as episiotomy, and immediate newborn care practices such as breastfeeding and skin-to-skin contact (Lothian, 2006).

The concept of birth plans is rooted in the principles of autonomy, informed choice, and shared decision-making. By encouraging women to articulate their preferences and expectations, birth

plans serve as a communication tool between patients and healthcare providers. This helps to reduce misunderstandings, align expectations, and foster a collaborative approach to care. Birth plans also empower women by giving them a sense of control over their childbirth experience, which is a critical determinant of satisfaction.

Empirical evidence suggests that the use of birth plans is associated with improved maternal experiences and outcomes. Studies have shown that women who use birth plans are more likely to feel involved in decision-making, experience lower levels of anxiety, and report higher satisfaction with their childbirth experience (Afshar et al., 2017). Furthermore, when healthcare providers acknowledge and attempt to adhere to birth plans, women are more likely to perceive their care as respectful and individualized.

However, the effectiveness of birth plans is influenced by several contextual factors, including healthcare system capacity, provider attitudes, and institutional policies. In some cases, birth plans may not be fully implemented due to medical emergencies, resource constraints, or lack of provider support. Additionally, there is evidence that some healthcare providers perceive birth plans as rigid or unrealistic, which may affect their willingness to engage with them (Lundgren et al., 2012). These challenges highlight the need to assess the practical effectiveness of birth plans within specific healthcare settings.

In Ghana, maternal health remains a priority, with significant efforts made to improve access to skilled delivery services. The introduction of policies such as the Free Maternal Healthcare Policy under the National Health Insurance Scheme has led to increased utilization of antenatal and delivery services. Current estimates suggest that approximately 74% of births in Ghana occur in health facilities (Ghana Statistical Service [GSS], 2018). Despite this progress, concerns persist regarding the quality of care and women's experiences during childbirth.

Studies conducted in Ghana have identified several factors influencing maternal satisfaction, including healthcare providers' attitudes, communication, waiting time, privacy, and availability of essential resources (Avortri, Beke, & Abekah-Nkrumah, 2011). Additionally, research in Northern Ghana has shown that women's involvement in decision-making and the level of emotional support received during labour significantly affect their satisfaction with maternal healthcare services (Yakubu et al., 2021).

Despite these findings, there is limited research on the use and effectiveness of birth plans in Ghanaian healthcare settings. In particular, little is known about how birth plans influence maternal satisfaction in district-level hospitals such as Sunyani Municipal Hospital. Given the growing emphasis on patient-centered care and respectful maternity services, it is important to explore whether birth plans can serve as an effective tool for improving maternal satisfaction in this context.

This study therefore seeks to assess the effectiveness of birth plans in improving maternal satisfaction among women at Sunyani Municipal Hospital.

1.2 Problem Statement

Over the years, Ghana has made considerable progress in improving access to maternal healthcare services, particularly through increased antenatal care attendance and facility-based deliveries. However, access alone does not guarantee quality care or positive health outcomes. Increasingly, attention has shifted toward evaluating the quality of care from the perspective of service users, particularly maternal satisfaction.

Maternal satisfaction is a critical component of healthcare quality because it influences women's health-seeking behavior, adherence to medical advice, and future utilization of healthcare services. Despite high antenatal care coverage in Ghana, there remains a noticeable

gap between antenatal attendance and facility-based delivery in some regions of sub-Saharan Africa. One of the contributing factors to this gap is dissatisfaction with previous childbirth experiences (Bohren et al., 2014).

Several studies have documented instances of poor communication, disrespectful care, lack of privacy, and limited involvement of women in decision-making during childbirth in many healthcare settings. These negative experiences not only affect maternal satisfaction but can also have long-term psychological effects, including fear of childbirth and reduced trust in healthcare systems.

Birth plans have been proposed as a strategy to address these challenges by promoting communication, enhancing patient autonomy, and facilitating shared decision-making. However, in many healthcare settings, including Ghana, birth plans are not widely utilized. Even where they are used, their implementation is often inconsistent due to factors such as healthcare provider attitudes, lack of training, time constraints, and institutional limitations.

At Sunyani Municipal Hospital, although maternal health services are widely utilized, there is limited empirical evidence on whether birth plans are being used effectively and whether they contribute to improved maternal satisfaction. Anecdotal reports suggest that some women may not be adequately involved in decisions regarding their care, and communication between healthcare providers and patients may not always be optimal.

The lack of data on the effectiveness of birth plans in this setting presents a significant gap in knowledge. Without such evidence, it is difficult to determine whether promoting birth plans as part of routine maternal care would lead to improved patient experiences and outcomes.

Therefore, this study aims to address this gap by assessing the effectiveness of birth plans in improving maternal satisfaction among women at Sunyani Municipal Hospital.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to assess the effectiveness of birth plans in improving maternal satisfaction among women at Sunyani Municipal Hospital.

1.3.2 Specific Objectives

The specific objectives of the study are:

1. To examine the level of awareness and utilization of birth plans among pregnant women attending Sunyani Municipal Hospital.
2. To evaluate the relationship between the use of birth plans and maternal satisfaction during childbirth.
3. To identify the factors influencing the effectiveness of birth plans in improving maternal satisfaction.

1.4 Research Questions

This study seeks to answer the following research questions:

1. What is the level of awareness and utilization of birth plans among women at Sunyani Municipal Hospital?
2. What is the relationship between the use of birth plans and maternal satisfaction during childbirth?
3. What factors influence the effectiveness of birth plans in improving maternal satisfaction?

1.5 Significance of the Study

This study holds significant academic, clinical, and policy relevance.

From an academic perspective, the study contributes to the existing body of knowledge on maternal healthcare by providing empirical evidence on the role of birth plans in improving maternal satisfaction within a Ghanaian context. Given the limited research in this area, particularly at the district hospital level, the findings will help fill an important gap in the literature.

From a clinical perspective, the study will provide valuable insights for healthcare providers, including midwives, nurses, and obstetricians. By understanding how birth plans influence maternal satisfaction, healthcare providers can adopt more patient-centered approaches, improve communication with patients, and enhance the overall quality of care.

From a policy perspective, the findings of this study will be useful to policymakers and healthcare administrators in designing interventions aimed at improving maternal healthcare services. Promoting the use of birth plans may align with national and global efforts to enhance respectful maternity care and improve patient experiences.

Furthermore, the study will empower pregnant women by highlighting the importance of active participation in their care. Encouraging the use of birth plans can help women express their preferences, make informed decisions, and have a more positive childbirth experience.

1.6 Organization of Chapters

This study is organized into three main chapters.

Chapter One provides the introduction to the study, including the background, problem statement, objectives, research questions, significance, and structure of the study.

Chapter Two presents a comprehensive review of relevant literature, focusing on key concepts such as maternal satisfaction, birth plans, and factors influencing their effectiveness.

Chapter Three outlines the methodology of the study, including the research design, study population, sampling techniques, data collection methods, data analysis procedures, ethical considerations, and limitations of the study.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter presents a comprehensive review of literature relevant to the study on the effectiveness of birth plans in improving maternal satisfaction. The purpose of this review is to situate the current study within existing academic discourse, identify gaps in knowledge, and provide a conceptual and empirical foundation for the research. The review draws on both theoretical perspectives and empirical studies from global and local contexts, with particular attention to maternal healthcare systems in low and middle-income countries.

Over time, the concept of maternal healthcare has undergone a significant transformation. Traditionally, maternal health interventions were primarily concerned with reducing maternal and neonatal mortality. However, contemporary approaches emphasize not only survival but also the quality of care and the subjective experiences of women during childbirth. This shift reflects the growing recognition that how care is delivered is just as important as the clinical outcomes achieved.

Central to this transformation is the concept of maternal satisfaction, which captures women's perceptions and evaluations of the care they receive. Maternal satisfaction is now widely regarded as a key indicator of healthcare quality, particularly in maternity services. It encompasses a range of dimensions, including emotional support, communication, respect, and involvement in decision-making. These dimensions align closely with the principles of respectful maternity care, which advocate for dignity, autonomy, and individualized care.

Within this evolving framework, birth plans have emerged as a practical tool for promoting patient-centered care. Originating in the 1980s as a response to the increasing medicalization of childbirth, birth plans were designed to empower women and enhance their participation in decisions regarding labour and delivery (Bell et al., 2022). Although initially associated with natural childbirth movements, birth plans have since evolved into flexible instruments that can be adapted to different clinical contexts.

Despite their potential benefits, the effectiveness of birth plans remains a subject of ongoing debate. While some studies report improved maternal satisfaction and better communication, others highlight challenges related to implementation, unrealistic expectations, and provider resistance. These mixed findings underscore the need for context-specific research to better understand how birth plans function in different healthcare environments.

This chapter is therefore organized around three key thematic areas derived from the study's specific objectives. The first section examines awareness and utilization of birth plans, the second explores the relationship between birth plans and maternal satisfaction, and the third analyzes the factors influencing the effectiveness of birth plans. Together, these sections provide a comprehensive understanding of the topic and establish a foundation for the current study.

2.2 Key Topics for First Specific Objective: Awareness and Utilization of Birth Plans

2.2.1 Concept and Evolution of Birth Plans

The concept of birth plans originated as part of a broader movement advocating for the humanization of childbirth. During the late twentieth century, concerns emerged regarding the increasing medicalization of childbirth, characterized by routine interventions such as continuous fetal monitoring, induction of labour, and cesarean sections. Critics argued that these practices often undermined women's autonomy and reduced childbirth to a purely clinical process.

In response, birth plans were introduced as a mechanism to restore women's agency and promote a more holistic approach to childbirth. According to Bell et al. (2022), birth plans were initially developed to facilitate communication between women and healthcare providers and to ensure that women's preferences were considered during labour and delivery. Over time, the scope of birth plans expanded to include a wide range of preferences and expectations.

A typical birth plan may include preferences related to pain management (such as the use of epidural anesthesia or non-pharmacological methods), labour positioning (e.g., upright or squatting positions), the presence of a birth companion, and decisions regarding interventions such as episiotomy or cesarean section. It may also address postpartum practices, including breastfeeding initiation and newborn care.

Importantly, contemporary literature emphasizes that birth plans should not be viewed as rigid directives but rather as flexible guides that facilitate dialogue and shared decision-making. This shift reflects an understanding that childbirth is inherently unpredictable and that flexibility is essential to accommodate clinical realities. Thus, the evolution of birth plans highlights a transition from prescriptive documents to collaborative tools aimed at enhancing communication and patient-centered care.

2.2.2 Awareness of Birth Plans among Pregnant Women

Awareness of birth plans is a critical prerequisite for their utilization. Without adequate knowledge of what birth plans are and how they function, pregnant women are unlikely to engage in birth planning or benefit from its potential advantages. Existing literature indicates that awareness levels vary widely across different regions and healthcare systems.

In high-income countries, awareness of birth plans is generally high due to structured antenatal education programs, widespread access to information, and proactive engagement by healthcare providers. Women in these settings are often encouraged to develop birth plans as part of routine antenatal care.

In contrast, awareness levels in low- and middle-income countries remain relatively low. Studies suggest that many pregnant women in these contexts are not adequately informed about their rights, options, or the concept of birth planning. This gap in knowledge can be attributed to several factors, including limited health education, high patient loads, and insufficient provider engagement.

In the Ghanaian context, antenatal care services are widely utilized; however, the extent to which these services incorporate education on birth planning is unclear. Research indicates that antenatal education often focuses on clinical aspects of pregnancy, with less emphasis on patient autonomy and decision-making. As a result, many women may not be aware of their ability to express preferences or participate actively in their care.

Furthermore, sociocultural factors play a significant role in shaping awareness. In some communities, decision-making during childbirth is traditionally dominated by healthcare providers or family members, which may discourage women from seeking information or

asserting their preferences. Literacy levels also influence awareness, as women with limited education may find it difficult to access or understand information about birth plans.

2.2.3 Utilization of Birth Plans in Clinical Practice

Utilization of birth plans refers to both the development of birth plans by pregnant women and their implementation during labour and delivery. While awareness is a necessary condition for utilization, it is not sufficient on its own. Effective utilization requires active engagement from both women and healthcare providers.

Empirical studies indicate that the use of birth plans is highly variable across healthcare settings. In some contexts, birth plans are integrated into routine antenatal care, with healthcare providers actively supporting women in their development and implementation. In such cases, birth plans serve as a valuable tool for enhancing communication and improving maternal satisfaction.

However, in many settings, including resource-constrained environments, the use of birth plans remains limited. Factors such as high patient-to-provider ratios, time constraints, and lack of standardized protocols hinder their integration into clinical practice. In such contexts, healthcare providers may prioritize clinical tasks over individualized care planning, reducing opportunities for meaningful engagement with patients.

Research further suggests that collaborative birth planning, where women and healthcare providers jointly develop the plan, is more effective than approaches where either party acts independently (Bell et al., 2022). Collaborative approaches ensure that birth plans are realistic, medically appropriate, and aligned with available resources.

In contrast, when birth plans are developed without provider input, they may include unrealistic expectations that cannot be met during labour. This mismatch between expectations and reality can lead to dissatisfaction and undermine the potential benefits of birth planning.

2.2.4 Barriers to Awareness and Utilization

The literature identifies several barriers that hinder the awareness and utilization of birth plans. These barriers operate at multiple levels, including the healthcare system, providers, and individual patients.

Healthcare system constraints represent a major challenge. In many low-resource settings, healthcare facilities are overburdened, with limited staff and high patient volumes. Under such conditions, healthcare providers may have insufficient time to engage in detailed discussions with patients or support the development of birth plans. Additionally, the absence of institutional policies or guidelines on birth planning further limits its adoption.

Provider attitudes also play a crucial role. Some healthcare providers perceive birth plans as unrealistic or unnecessary, particularly in settings where clinical priorities dominate. This perception may lead to resistance or lack of engagement, thereby reducing the effectiveness of birth plans.

Sociocultural factors constitute another important barrier. In some societies, women may defer decision-making to healthcare providers or family members due to cultural norms. This can limit their willingness or ability to develop and use birth plans.

Educational barriers further compound these challenges. Women with limited education may lack the knowledge or confidence to engage in birth planning. They may also face difficulties in understanding medical information or communicating their preferences effectively.

These barriers highlight the need for targeted interventions that address both systemic and individual-level factors to promote the effective use of birth plans.

2.3 Key Topics for Second Specific Objective: Relationship Between Birth Plans and Maternal Satisfaction

2.3.1 Concept of Maternal Satisfaction

Maternal satisfaction is a complex and multidimensional construct that reflects women's perceptions of the care they receive during childbirth. It encompasses both objective and subjective elements, including clinical outcomes, interpersonal interactions, and emotional experiences.

One of the most important dimensions of maternal satisfaction is the quality of interpersonal relationships between women and healthcare providers. Respectful, empathetic, and supportive interactions are consistently associated with higher levels of satisfaction. Conversely, negative interactions, such as neglect or disrespect, can significantly undermine women's experiences.

Communication is another critical determinant. Clear and timely information enables women to understand what is happening during labour and make informed decisions. Lack of communication, on the other hand, can lead to confusion, anxiety, and dissatisfaction.

Emotional support, including reassurance and encouragement, also plays a vital role. Women who feel supported during labour are more likely to have positive experiences, regardless of the clinical outcome.

Participation in decision-making is equally important. Women who are actively involved in decisions regarding their care tend to feel more empowered and satisfied.

2.3.2 Theoretical Link Between Birth Plans and Satisfaction

The relationship between birth plans and maternal satisfaction can be understood through the lens of patient-centered care and empowerment theory. These frameworks emphasize the importance of respecting patients' preferences, involving them in decision making, and providing individualized care.

Birth plans operationalize these principles by creating a structured mechanism for expressing preferences and facilitating communication. Through birth plans, women are encouraged to reflect on their expectations, seek information, and engage in discussions with healthcare providers.

This process enhances women's sense of control, which is a key determinant of satisfaction. When women feel that they have a voice in their care and that their preferences are respected, they are more likely to perceive their experience positively.

2.3.3 Empirical Evidence Supporting Positive Outcomes

Empirical studies provide substantial evidence supporting the positive impact of birth plans on maternal satisfaction. For instance, Mirghafourvand et al. (2019) found that women who used birth plans reported higher levels of satisfaction and a greater sense of control during childbirth.

Similarly, Ahmadpour et al. (2022) reported that birth plans contributed to reduced anxiety and improved psychological well-being. These findings suggest that birth plans not only influence clinical outcomes but also enhance the emotional and psychological aspects of childbirth.

Furthermore, Kohan et al. (2023) found that birth planning was associated with improved maternal-infant outcomes, including increased rates of early breastfeeding and reduced medical interventions. These outcomes further reinforce the value of birth plans as a tool for improving overall maternal healthcare.

2.3.4 Contradictory Evidence

Despite the positive findings, some studies have reported less favorable outcomes. Afshar et al. (2018) found that women with birth plans sometimes reported lower satisfaction levels, particularly when their expectations were not met.

This contradiction highlights the complexity of the relationship between birth plans and satisfaction. While birth plans have the potential to enhance experiences, they may also create expectations that are difficult to fulfill in practice. When these expectations are unmet, women may feel disappointed or dissatisfied.

2.3.5 Importance of Expectation Management

Expectation management emerges as a critical factor in determining the effectiveness of birth plans. Satisfaction is closely linked to the extent to which expectations align with actual experiences.

Healthcare providers play a key role in managing expectations by providing realistic information and emphasizing the need for flexibility. When women understand that birth plans are guidelines rather than guarantees, they are more likely to adapt to changing circumstances and maintain a positive outlook.

2.4 Factors Influencing the Effectiveness of Birth Plans

2.4.1 Healthcare Provider Attitudes and Practices

Healthcare providers are central to the successful implementation of birth plans. Their attitudes toward patient involvement and willingness to engage in shared decision-making significantly influence outcomes.

Positive provider attitudes foster trust, encourage communication, and enhance the likelihood that birth plans will be respected. In contrast, negative attitudes or resistance can undermine the effectiveness of birth plans.

2.4.2 Communication and Shared Decision-Making

Effective communication is essential for translating birth plans into practice. It ensures that healthcare providers understand patients' preferences and can respond appropriately.

Shared decision-making further strengthens this process by involving both parties in the decision-making process. This collaborative approach enhances trust and improves satisfaction.

2.4.3 Health System Factors

Health system characteristics, including resource availability, staffing levels, and institutional policies, play a significant role in shaping the effectiveness of birth plans.

In resource-limited settings, constraints such as staff shortages and inadequate infrastructure may limit the ability to implement birth plans fully.

2.4.4 Sociocultural Factors

Cultural norms and beliefs influence women's expectations and behaviors. In some contexts, women may be reluctant to assert their preferences due to cultural expectations of deference to authority.

2.4.5 Educational Level and Awareness

Education enhances women's ability to understand medical information, communicate effectively, and participate in decision-making. As such, it is a key determinant of birth plan utilization and effectiveness.

2.4.6 Flexibility of Birth Plans

Flexibility is essential for accommodating the unpredictable nature of childbirth. Birth plans that allow for adjustments are more likely to be successfully implemented and result in positive experiences.

2.5 Conclusion

The literature demonstrates that birth plans have significant potential to improve maternal satisfaction by enhancing communication, promoting autonomy, and aligning care with women's preferences. However, their effectiveness is influenced by multiple factors, including provider attitudes, health system capacity, and sociocultural context.

Importantly, there remains a gap in research within the Ghanaian context, particularly at district-level facilities such as Sunyani Municipal Hospital. This underscores the relevance and necessity of the current study.

CHAPTER THREE

METHODOLOGY

3.1 Background of the Study Area

The study will be conducted at Sunyani Municipal Hospital, located in the Bono Region of Ghana. Sunyani serves as the administrative and commercial capital of the region and is a major urban center that attracts populations from surrounding districts for economic, social, and healthcare services. The municipality is characterized by a heterogeneous population comprising urban, peri-urban, and rural dwellers, which results in diverse healthcare needs, utilization patterns, and health-seeking behaviors (Ghana Statistical Service [GSS], 2018).

Sunyani Municipal Hospital is a government-owned health facility that plays a pivotal role in the delivery of primary and secondary healthcare services within the municipality. The hospital operates under the Ghana Health Service and adheres to national health policies, including the Free Maternal Healthcare Policy under the National Health Insurance Scheme. It provides comprehensive maternal and child health services such as antenatal care (ANC), skilled labour and delivery services, postnatal care (PNC), family planning services, and emergency obstetric and neonatal care (EmONC), which are essential for improving maternal and neonatal outcomes (Ministry of Health Ghana, 2020).

The maternity unit of the hospital is staffed by trained midwives, nurses, and medical officers who provide skilled birth attendance in line with WHO recommendations for quality maternal care. The facility manages both uncomplicated and complicated deliveries and receives referrals from peripheral health facilities such as Community-based Health Planning and

Services (CHPS) compounds and health centers. This referral role contributes to a high patient load, which is advantageous for research as it ensures a sufficiently large and diverse sample population (WHO, 2016).

The hospital records a significant number of deliveries annually, making it an appropriate setting for assessing maternal satisfaction and the effectiveness of birth plans. High service utilization increases variability in patient experiences, which is critical for examining factors such as communication, provider attitudes, and decision-making processes during childbirth.

Furthermore, Sunyani Municipal Hospital reflects the operational realities of many district-level healthcare facilities in Ghana, including challenges such as limited human resources, high workload, infrastructural constraints, and variability in quality of care. These contextual factors are important in understanding how birth plans are implemented and how they influence maternal satisfaction in real-world settings (Bohren et al., 2015).

The selection of this study area is therefore justified by its accessibility, high client flow, and representativeness of similar healthcare settings in Ghana. This enhances the external validity and applicability of the study findings.

3.2 Study Design and Type

This study will employ a descriptive cross-sectional study design using a quantitative research approach.

A cross-sectional study design involves the collection of data from a defined population at a single point in time. This design allows researchers to assess the prevalence of specific characteristics and examine relationships between variables without requiring follow-up over time (Levin, 2006).

The choice of a cross-sectional design is appropriate for several reasons. First, the study aims to assess the current level of awareness and utilization of birth plans as well as maternal satisfaction, which are variables that can be measured effectively at a single point in time. Second, the study seeks to examine associations between variables, rather than establish causality, making the cross-sectional approach suitable (Setia, 2016).

Cross-sectional designs are widely used in maternal health research because they are cost-effective, time-efficient, and capable of capturing real-time experiences of postpartum women. Studies conducted in similar contexts have successfully used this design to evaluate maternal satisfaction and determinants of quality care (Srivastava et al., 2015).

The study will adopt a quantitative approach, which involves the collection and analysis of numerical data using structured instruments. This approach allows for:

- a) Objective measurement of maternal satisfaction using standardized scales
- b) Statistical comparison between groups (e.g., users and non-users of birth plans)
- c) Identification of significant predictors of maternal satisfaction

Quantitative methods enhance the reliability, objectivity, and generalizability of findings, making them suitable for this study (Creswell, 2014).

3.3 Study Population

The study population will consist of postnatal women who have delivered at Sunyani Municipal Hospital during the study period.

The selection of postnatal women is based on the premise that maternal satisfaction is best assessed shortly after childbirth when the experience is still recent. Research indicates that

interviewing women within the immediate postpartum period reduces recall bias and improves the accuracy of reported experiences (Hodnett, 2002).

Inclusion Criteria

Women will be included if they meet the following criteria:

- a) Women who have delivered at Sunyani Municipal Hospital during the study period
- b) Women who are within 24–72 hours postpartum

This timeframe is widely used in maternal satisfaction studies because it balances recovery from childbirth with accurate recall of events (Srivastava et al., 2015).

- a) Women who are mentally and physically stable

Participants must be capable of engaging in an interview without discomfort or distress, ensuring data reliability.

- a) Women who provide informed consent

Participation must be voluntary, consistent with ethical research principles.

a) Exclusion Criteria

- b) Women who are critically ill
- c) Women with communication difficulties
- d) Women with severe obstetric or psychological complications

These exclusions are necessary to ensure participant safety and data validity, as such conditions may influence responses independently of the variables being studied (Bohren et al., 2015).

3.4 Sample Size and Sampling Technique

3.4.1 Sample Size Determination

The sample size will be determined using the single population proportion formula:

$$n = \frac{Z^2 \cdot p(1 - p)}{d^2}$$

Where:

- $Z = 1.96$ (95% confidence level)
- $p = 0.5$ (assumed proportion due to lack of prior data)
- $d = 0.05$ (margin of error)

$$n = \frac{(1.96)^2 \cdot 0.5(1 - 0.5)}{(0.05)^2} = 384$$

Adding a 10% non-response rate:

$$n = 384 + 38 = 422$$

The use of 50% prevalence ensures maximum sample size and statistical power when prior estimates are unavailable (Naing, Winn, & Rusli, 2006).

3.4.2 Sampling Technique

A **systematic random sampling technique** will be employed.

This method involves selecting participants at regular intervals from a sampling frame. It is appropriate for hospital-based studies where patients present in a continuous flow (Etikan & Bala, 2017).

The process will involve:

- a) Estimating total eligible participants (N)
- b) Calculating sampling interval $k = N/n$
- c) Randomly selecting the first participant
- d) Selecting every k th participant

This approach ensures representativeness while minimizing selection bias.

3.5 Study Variables

Dependent Variable

Maternal satisfaction

Maternal satisfaction will be measured using a Likert scale assessing multiple domains such as communication, respect, involvement in decision-making, and facility environment. This multidimensional approach aligns with established frameworks for measuring healthcare quality (Donabedian, 1988).

Independent Variables

Awareness of birth plans

Refers to knowledge of birth planning concepts and practices. Awareness influences utilization and participation in care (Afshar et al., 2017).

Utilization of birth plans

Refers to whether a woman developed and used a birth plan and whether it was implemented.

Socio-demographic factors

Age, education, and occupation influence access to information and healthcare utilization.

Obstetric factors

Parity, delivery mode, and complications affect childbirth experiences.

Health system factors

Include provider attitude, communication, waiting time, and facility conditions, all of which significantly influence satisfaction (Bohren et al., 2015).

3.6 Data Collection Tool and Technique

Data will be collected using a structured questionnaire administered via face-to-face interviews.

This method is appropriate for ensuring comprehension among participants with varying literacy levels and allows clarification of questions (Creswell, 2014).

The questionnaire will include:

- a) Socio-demographic data
- b) Obstetric history
- c) Birth plan awareness and utilization
- d) Maternal satisfaction scale

The instrument will be adapted from validated tools such as the Labour and Delivery Satisfaction Index, ensuring reliability and validity (Hodnett, 2002).

Research assistants will be trained to ensure:

- a) Standardization of data collection
- b) Minimization of interviewer bias
- c) Ethical compliance

3.7 Data Processing and Analysis

Data will be analyzed using **SPSS**.

Data Processing

- a) Data cleaning to remove inconsistencies
- b) Coding for statistical analysis
- c) Data entry into SPSS

Data Analysis

Descriptive statistics

Used to summarize data through frequencies, percentages, means, and standard deviations.

Inferential statistics

- a) Chi-square tests to assess associations
- b) Logistic regression to identify predictors

A significance level of **$p < 0.05$** will be used (Field, 2013).

3.8 Limitations of the Study

The cross-sectional design limits causal inference (Setia, 2016).

Self-reported data may introduce bias (Bohren et al., 2015).

Single-site study limits generalizability.

Operational constraints may affect data collection.

3.9 Ethical Considerations

Ethical approval will be obtained from relevant authorities.

Participants will:

- a) Provide informed consent
- b) Be assured of confidentiality
- c) Participate voluntarily
- d) Remain anonymous

These principles align with international ethical standards for human research (WHO, 2016).

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